At the earliest time shown in the diagram (bottom horizontal slice), the star, with one spatial dimension absent, is the interior of a large circle; if the missing dimension were restored, the star would be the interior of a large sphere. At a later time (second slice), the star has shrunk; it is now the interior of a smaller circle. At a still later time, the star passes through its critical circumference, and still later it shrinks to zero circumference, creating there a *singularity* in which, according to general relativity, the star is crunched out of existence. We shall not discuss the details of this singularity until Chapter 13, but it is crucial to know that it is a completely different thing from the "Schwarzschild singularity" of which physicists spoke from the 1920s through the 1950s. The "Schwarzschild singularity" was their ill-conceived name for the critical circumference or for a black hole; this "singularity" is the object that resides at the black hole's center.

The black hole itself is the region of spacetime that is shown black in the diagram, that is, the region inside the critical circumference and to the future of the imploding star's surface. The hole's surface (its *horizon)* is at the critical circumference.

Also shown in the diagram are the world lines (trajectories through spacetime) of some particles attached to the star's surface. As one's eye travels upward in the diagram (that is, as time passes), one sees these world lines move in closer and closer to the center of the star (to the central axis of the diagram). This motion exhibits the star's shrinkage with time.

Of greatest interest are the world lines of four photons (four particles of light). These photons are the analogues of the signal balls in the parable of the ants. Photon A is emitted outward from the star's surface at the moment when the star begins to implode (bottom slice). It travels outward with ease, to larger and larger circumferences, as time passes (as one's eye travels upward in the diagram). Photon B, emitted shortly before the star reaches its critical circumference, requires a long time to escape; it is the analogue of signal ball number 14.999 in the parable of the ants. Photon C, emitted precisely at the critical circumference,. remains always there, just like signal ball number 15 And photon D, emitted from inside the critical circumference (inside the black hole), never escapes; it gets pulled into the singularity by the hole's intense gravity, just like signal ball 15.001.